

From Particles to Waves

Towards Coupling Dynamic
Snow Modelling and Spectral
Elements to Detect
Avalanche Breakoff in DAS
Data

Salome Bachmann

August 14, 2026



Why Model Crack Propagation in Snow? - Relevance

- 22 fatalities by avalanche in Switzerland every year (SLF, 2025)
- Avalanches are danger for human life but also infrastructure (Schweizer et al., 2003)
- Understanding processes associated with avalanches important
- Avalanche release because of loading, leading to crack propagation in a weak layer (Gaume et al., 2017)

Relevance

- Crack propagation at different speeds: sub-Rayleigh (below v_s of material) and Supershear (above v_s) (Trottet et al., 2022), transition between two regimes possible (Bergfeld et al., 2025)
- Fast crack propagation $\rightarrow$ critical crack length for release (Gaume et al., 2015) is reached faster
- Distributed Acoustic Sensing (DAS) already used to monitor avalanches in Switzerland (Edme et al., 2023; Paitz et al., 2023) and Norway (Turquet et al., 2024)
- No knowledge if crack propagation associated with breakoff can be seen in DAS data

Overview

- **Goal:** Couple dynamic snow modelling with SE to model crack propagation in snow
- Created 3 different models and 2 scenarios for crack propagation in snow using SALVUS
- Two pure spectral element (SE) models, one combined with data from material point method (MPM) simulations

Nomenclature & Abbreviations

- **SE, FE, FEM**: Spectral elements, Finite Element (Method), done in SALVUS
- **MPM**: Material Point Method Numerical Modeling
- **PST**: Propagation Saw Test
- **F-k**: Frequency-Wavenumber Plot
- **DAS**: Distributed Acoustic Sensing (Fibre-Optics)
- **STF**: Source-time function (the wavelet fired by SALVUS)

Research Questions

- What FE modelling framework is required to simulate DAS measurements of crack propagation in snow, and what are the key assumptions and limitations?
- What physical phenomena are observed in a two-layer model with a simple STF and how do they differ for sub-Rayleigh versus Supershear crack propagation?
- To what extent do the results change if source wavelets from a MPM simulation of a PST are injected?
- How could the phenomena observed be used to distinguish crack propagation regimes in real-world DAS data and aid in detecting snow slab avalanche breakoff?

Simple Model I

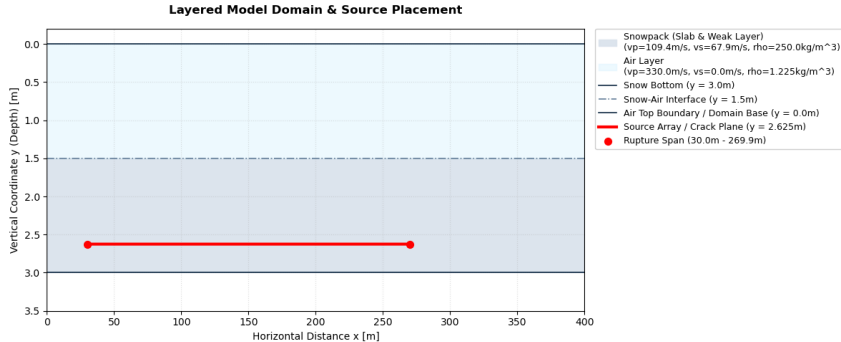


Figure: Two-layer model setup in SALVUS. Colors show materials, material types and properties in legend. Source line extent in red.

Simple Model II

- Set up to test custom STF in SALVUS, see physical effects on output
- Sources spaced over 240m in 10cm intervals
- 20Hz central frequency Ricker wavelet
- Source Wavelet in xx (1), xy (1) and yy (-1) direction
- Time delay of wavelet calculated based on propagation speed: sub-Rayleigh speed is $0.6v_s$, Supershear $1.6v_s$ (Trottet et al., 2022)

Results

- SALVUS Output: Displacement field u
- $\frac{\partial u_i}{\partial t} \rightarrow$ velocity in x and y
- $\frac{\partial u_i}{\partial j} \rightarrow$ strain in x and y ϵ
- $\frac{\partial^2 u_i}{\partial t \partial t} \rightarrow$ strain rate $\dot{\epsilon}$
- Simulation of DAS data from ϵ , $\dot{\epsilon}$ using Pyber Gauge length class by Noe, 2025, gauge length according to Turquet et al., 2024
- Spate-time plots of strain, strain rate and velocity at different receiver depths
- Frequency-wavenumber plots with theoretical speeds and dispersion curves
- Animations of velocity wavefields

Simple Sub-Rayleigh: Strain and Strain Rate

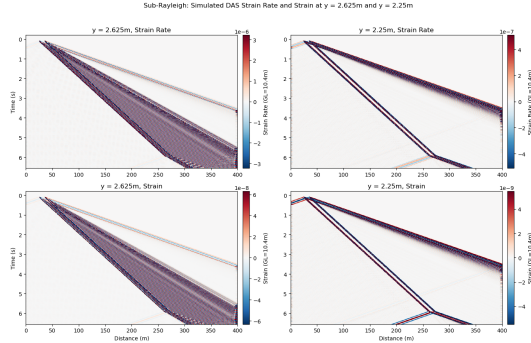
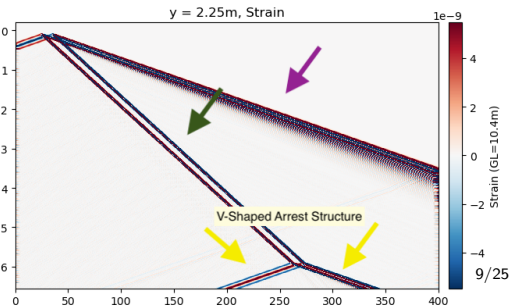
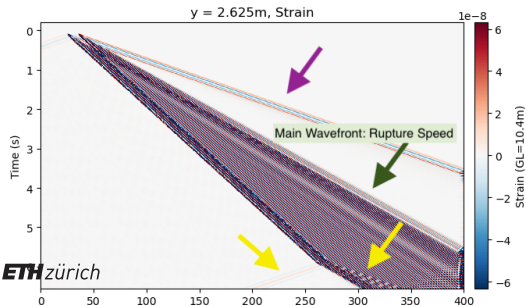
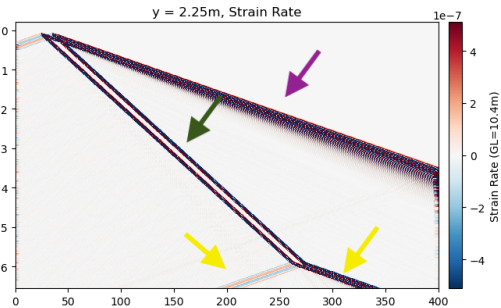
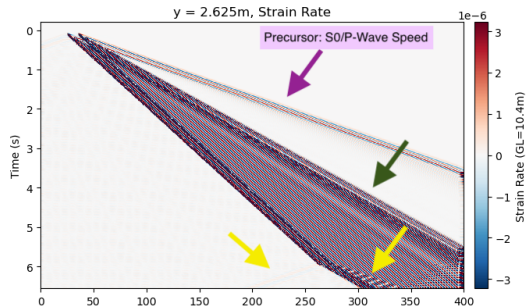


Figure: Horizontal Strain and strain rate for sub-Rayleigh crack propagation in simple two-layer model. Plotted at crack depth (2.625m) and at the snow bottom (2.25m). Gauge length is 10.4m.

Sub-Rayleigh: Simulated DAS Strain Rate and Strain at $y = 2.625\text{m}$ and $y = 2.25\text{m}$



Simple Sub-Rayleigh: Strain and Strain Rate

- Precursor emitted by all sources but they interfere (Madariaga, 1976), as shown by wider source spacing
- Slope of line close to p-wave speed in medium: S0 wave bounded by plate wave speed limit (“Waves in Plates”, 2014)
- S0/P-wave precursor observed in field data by Herwijnen and Schweizer, 2011
- Oscillatory behavior of both wavefronts (polarity flip at adjacent timesteps)
- Significant weakening of amplitude between two depths

Simple Supershear: Strain and Strain Rate

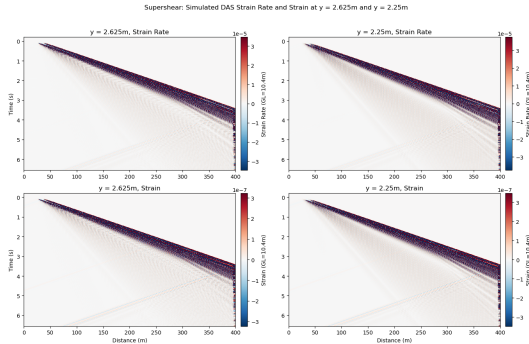
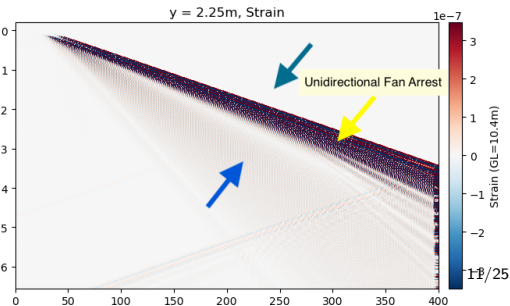
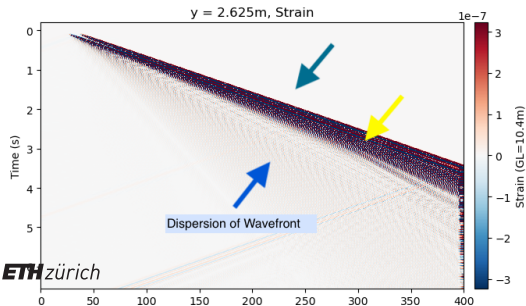
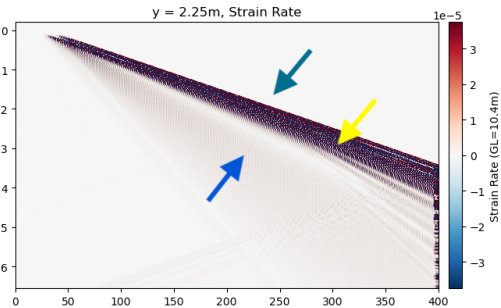
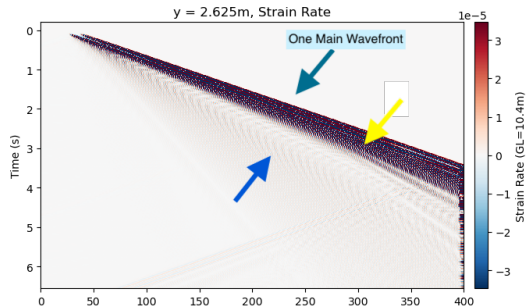


Figure: Horizontal Strain and strain rate for Supershear crack propagation in simple two-layer model. Plotted at crack depth (2.625m) and at the snow bottom (2.25m). Gauge length is 10.4m.

Supershear: Simulated DAS Strain Rate and Strain at $y = 2.625\text{m}$ and $y = 2.25\text{m}$



Simple Supershear: Strain and Strain Rate

- Dispersion and broadening of main wavefront (Aki & Richards, 1981)
- Oscillatory wavefront
- At $x=270\text{m}$, unidirectional fan-like arrest front: rupture outruns vs, so arrest energy is only propagated forward (Madariaga, 1977)
- S0 precursor can be seen in traces
- Supershear larger amplitude compared to sub-Rayleigh

Simple model: Lamb Wave Identification I

- Polarity oscillation in all plots: indicative of Lamb waves
- Plotting dispersion curves over f-k spectra: energy is concentrated at symmetric (S0) and antisymmetric (A0) mode
- Antisymmetric mode gets excited preferentially when source is applied normal to midplane of material (Achenbach, 1973; Lamb, 1917; "Waves in Plates", 2014), so in this case yy
- Confirmed by animations of wavefield:

Simple model: Lamb Wave Identification II

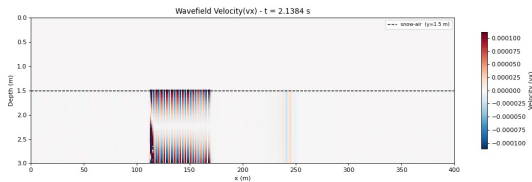


Figure: Animation still for Sub-Rayleigh Vx

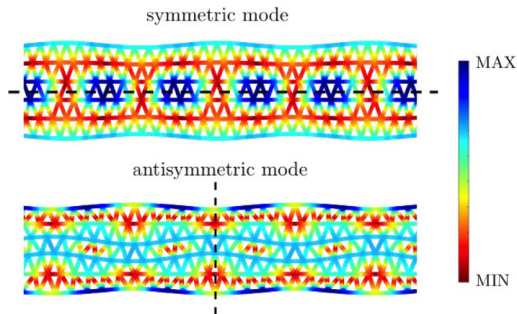


Figure: Lamb wave modes intensity, taken from Carta et al., 2023

MPM-Guided Model I

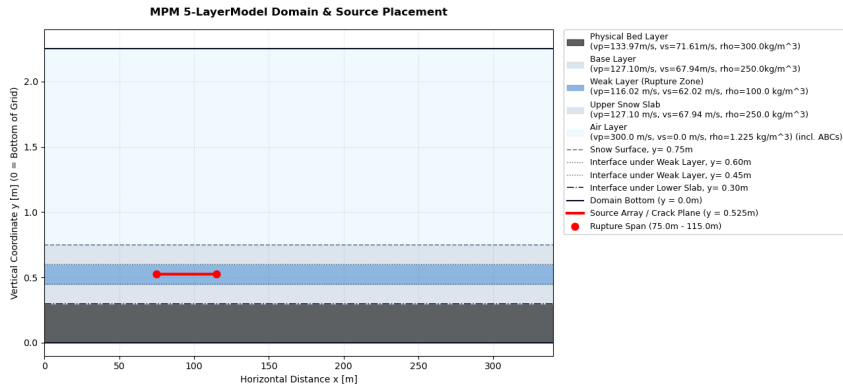


Figure: Five-layer model setup in SALVUS. Colors show materials, material types and properties in legend. Source line extent in red.

MPM-Guided Model II

- Source-time functions obtained from MPM-simulation of PST by Trottet et al., 2022
- Source Wavelet in all three directions, but irregular and different for every source point

MPM-Guided: Strain and Strain Rate

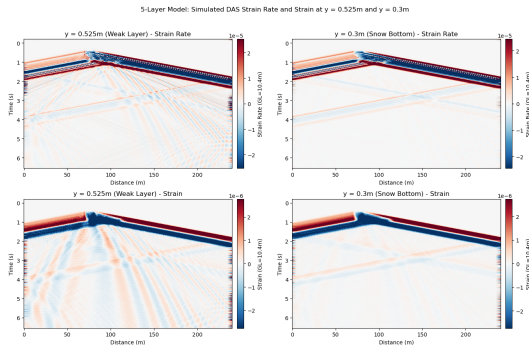
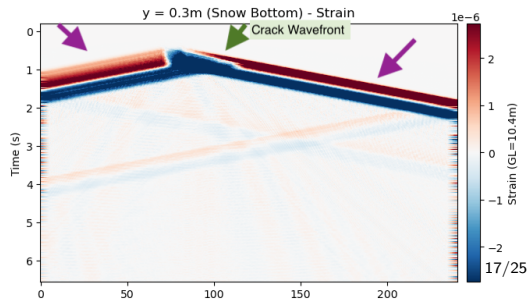
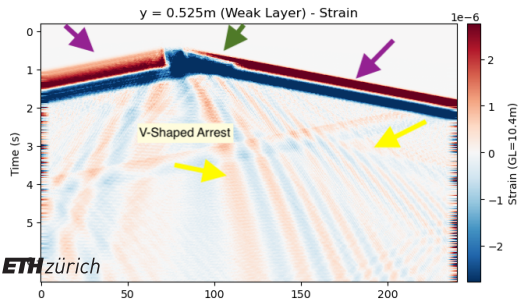
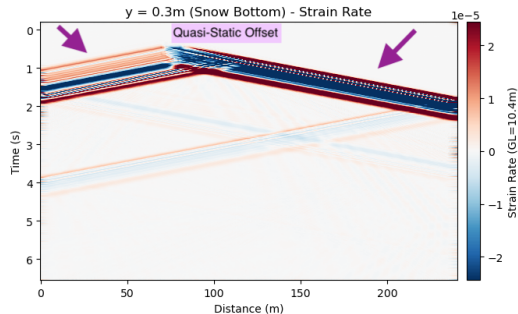
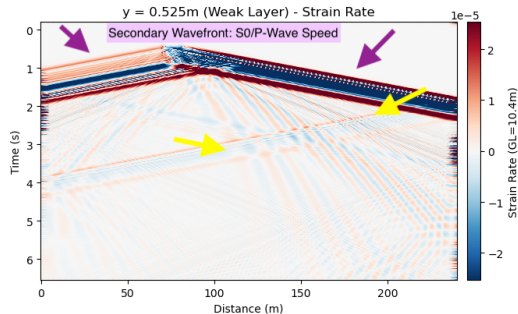


Figure: Horizontal Strain and strain rate for MPM-guided model. Plotted at crack depth (0.525) and at the snow bottom (0.3m). Gauge length is 10.4m.

5-Layer Model: Simulated DAS Strain Rate and Strain at $y = 0.525\text{m}$ and $y = 0.3\text{m}$



Quasi-Static Offset

- Static field represents permanent deformation after sources have fired (Archambeau, 1968; Zhu & Rivera, 2002)
- MPM STFS are not symmetric: contrary to Ricker wavelets, they do not integrate to 0 over time → net displacement of material (Aki & Richards, 1981; Zhu & Rivera, 2002)
- Cumulative integral of MPM STFs is negative: collapse of the weak layer, as observed by (Gaume et al., 2018; Heierli et al., 2008; Heierli et al., 2003; Trottet et al., 2022)
- All components of MPM STF (directionality) indicate mixed-mode anticrack instead of pure opening or shearing (mode I, II or III)

MPM-Guided Model: Lamb Wave Identification

- Like in simple model: oscillatory wavefronts $\rightarrow$ A0 mode

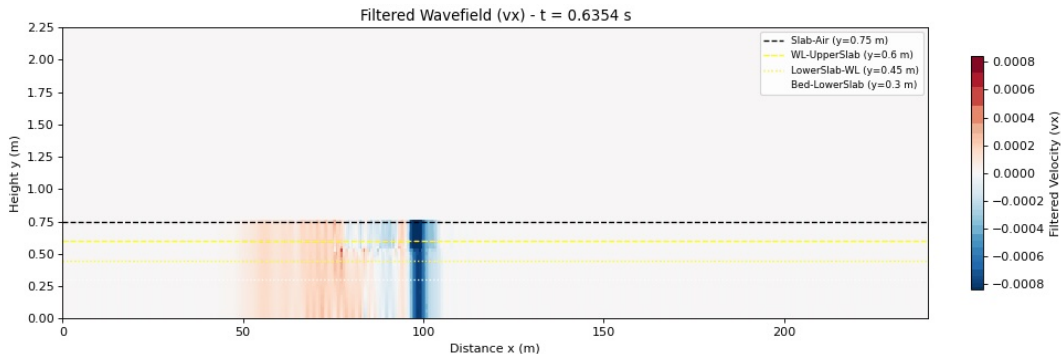
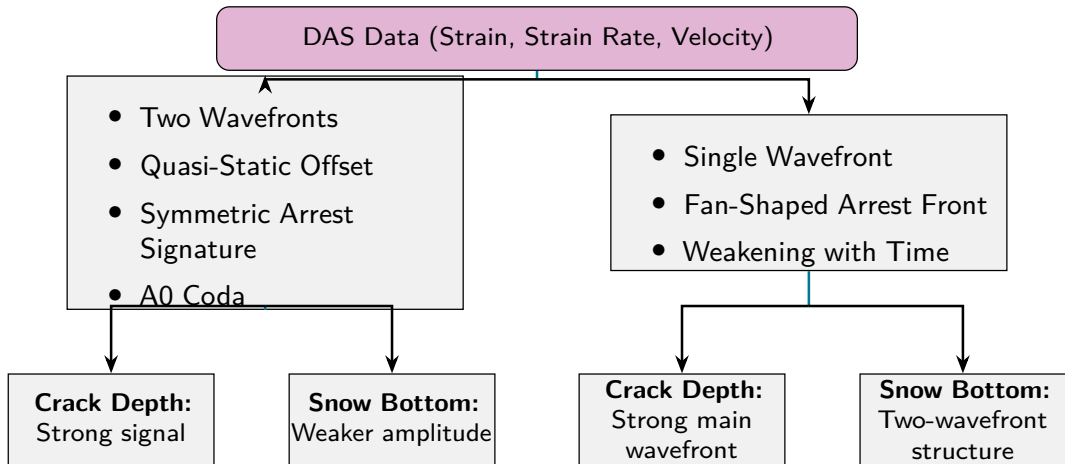


Figure: Animation still for MPM V_x

Summary of Results & Discussion I

- S0 Precursor observed in all Strain & Strain Rate Plots, as well as in traces
- Main wavfront in all cases is A0 $\rightarrow$ source normal to slab midplane
- Different rupture arrest structures depending on propagation speed
- Quasi-static wavefield for sources that do not integrate to 0 over time
- Weakening of amplitude only in simple case $\rightarrow$ also thicker slab

Implications for DAS field data I



Implications for DAS field data II

Figure: Graphical Representation to Identify Different Signatures in DAS

- Identification of rupture speed via number of wavefronts & arrest structure
- Precursors in sub-Rayleigh: S0 and Quasi-Static offset
- Consistent with observations by Herwijnen and Schweizer, 2011
- Structures identified as being geometry-independent
- Signals identified as being larger than instrument noise of Silixa iDAS used by Paitz et al., 2023
- Supershear signals larger in amplitude: possibly false conclusions if transition happens
- Identification flowchart:

Outlook

- Add topography to models
- Verify Supershear effects for more complex STFs as well
- Inhomogeneous, multi-layered snowpack
- Transitional propagation regimes
- Comparison to field data → amplitude of signals compared to noise
- 3D model

Answer to Research Questions - Conclusions

- **What physical phenomena are observed in a two-layer model with a simple STF and how do they differ for sub-Rayleigh versus Supershear crack propagation?** Rupture-arrest signatures depending on propagation speed, S0 precursor, A0 main wavefront for both propagation regimes. Changes in amplitude
- **To what extent do the results change if source wavelets from a MPM simulation of a propagation saw test (PST) are injected?** Verification of sub-Rayleigh S0 precursor, A0 mode excitation, changes because appearance of quasi-static offset
- **How could the phenomena observed be used to distinguish crack propagation regimes in real-world DAS data and aid in detecting snow slab avalanche breakoff?** Precursors (S0 and quasi-static offset) used as "early" warning, identification of propagation speed using arrest structures, possibility of identifying transitions by number of wavefronts

Questions?

Answer to Research Questions - Conclusions

- **What physical phenomena are observed in a two-layer model with a simple STF and how do they differ for sub-Rayleigh versus Supershear crack propagation?** Rupture-arrest signatures depending on propagation speed, S0 precursor, A0 main wavefront for both propagation regimes. Changes in amplitude
- **To what extent do the results change if source wavelets from a MPM simulation of a propagation saw test (PST) are injected?** Verification of sub-Rayleigh S0 precursor, A0 mode excitation, changes because appearance of quasi-static offset
- **How could the phenomena observed be used to distinguish crack propagation regimes in real-world DAS data and aid in detecting snow slab avalanche breakoff?** Precursors (S0 and quasi-static offset) used as "early" warning, identification of propagation speed using arrest structures, possibility of identifying transitions by number of wavefronts

Bibliography I



Achenbach, J. (1973). *Wave Propagation in Elastic Solids*. Elsevier.



Aki, K., & Richards, P. G. (1981). K. Aki & P. G. Richards 1980. Quantitative Seismology, Theory and Methods. Volume I: 557 pp., 169 illustrations. Volume II: 373 pp., 116 illustrations. San Francisco: Freeman. Price: Volume I, U.S. 35.00. ISBN 0 7167 1058 7 (Vol. I), 0 7167 1059 5 (Vol. II).. *Geological Magazine*, 118(2), 208–208. <https://doi.org/10.1017/S0016756800034439>



Archambeau, C. B. (1968). General theory of elastodynamic source fields. *Reviews of Geophysics*, 6(3), 241–288. <https://doi.org/10.1029/RG006i003p00241>



Bergfeld, B., Gaume, J., Bobillier, G., Pellet, A., Schweizer, J., & van Herwijnen, A. (2025). Signatures of the sub-Rayleigh to supershear fracture transition in snow avalanche experiments. *Nature Communications*, 16(1), 11153. <https://doi.org/10.1038/s41467-025-65825-6>

Bibliography II



Carta, G., Nieves, M. J., & Brun, M. (2023). Lamb waves in discrete homogeneous and heterogeneous systems: Dispersion properties, asymptotics and non-symmetric wave propagation. *European Journal of Mechanics - A/Solids*, 100, 104695.
<https://doi.org/10.1016/j.euromechsol.2022.104695> ↗



Edme, P., Paitz, P., Walter, F., van Herwijnen, A., & Fichtner, A. (2023, February). Fiber-optic detection of snow avalanches using telecommunication infrastructure.
<https://doi.org/10.48550/arXiv.2302.12649> ↗



Gaume, J., Gast, T., Teran, J., van Herwijnen, A., & Jiang, C. (2018). Dynamic anticrack propagation in snow. *Nature Communications*, 9(1), 3047.
<https://doi.org/10.1038/s41467-018-05181-w> ↗



Gaume, J., van Herwijnen, A., Chambon, G., Birkeland, K. W., & Schweizer, J. (2015). Modeling of crack propagation in weak snowpack layers using the discrete element method. *The Cryosphere*, 9(5), 1915–1932.
<https://doi.org/10.5194/tc-9-1915-2015> ↗

Bibliography III

-  Gaume, J., van Herwijnen, A., Chambon, G., Wever, N., & Schweizer, J. (2017). Snow fracture in relation to slab avalanche release: Critical state for the onset of crack propagation. *The Cryosphere*, 11(1), 217–228.
<https://doi.org/10.5194/tc-11-217-2017> 
-  Heierli, J., Gumbsch, P., & Zaiser, M. (2008). Anticrack Nucleation as Triggering Mechanism for Snow Slab Avalanches. *Science*, 321(5886), 240–243.
<https://doi.org/10.1126/science.1153948> 
-  Heierli, J., van Herwijnen, A., Gumbsch, P., & Zaiser, M. (2003). Anticracks: A new theory of fracture initiation and fracture propagation in snow.
-  Herwijnen, A. V., & Schweizer, J. (2011). Seismic sensor array for monitoring an avalanche start zone: Design, deployment and preliminary results. *Journal of Glaciology*, 57(202), 267–276. <https://doi.org/10.3189/002214311796405933> 
-  Lamb, H. (1917). On waves in an elastic plate. *Proceedings of the Royal Society of London. Series A, Containing Papers of a Mathematical and Physical Character*, 93(648), 114–128. <https://doi.org/10.1098/rspa.1917.0008> 

Bibliography IV



Madariaga, R. (1976). Dynamics of an expanding circular fault. *Bulletin of the Seismological Society of America*, 66(3), 639–666.

<https://doi.org/10.1785/BSSA0660030639>



Madariaga, R. (1977). High-frequency radiation from crack (stress drop) models of earthquake faulting. *Geophysical Journal International*, 51(3), 625–651.

<https://doi.org/10.1111/j.1365-246X.1977.tb04211.x>



Noe, S. (2025). Frontiers of full-waveform seismology: From global tomography to distributed and integrated fibre-optic strain sensing.

<https://doi.org/10.3929/ETHZ-C-000788455>



Paitz, P., Lindner, N., Edme, P., Huguenin, P., Hohl, M., Sovilla, B., Walter, F., & Fichtner, A. (2023). Phenomenology of Avalanche Recordings From Distributed Acoustic Sensing. *Journal of Geophysical Research: Earth Surface*, 128(5), e2022JF007011.

<https://doi.org/10.1029/2022JF007011>



Schweizer, J., Bruce Jamieson, J., & Schneebeli, M. (2003). Snow avalanche formation. *Reviews of Geophysics*, 41(4). <https://doi.org/10.1029/2002RG000123>

Bibliography V



SLF. (2025). Avalanche Victims since 1936 - Long Term Statistics.



Trottet, B., Simenhois, R., Bobillier, G., Bergfeld, B., van Herwijnen, A., Jiang, C., & Gaume, J. (2022). Transition from sub-Rayleigh anticrack to supershear crack propagation in snow avalanches. *Nature Physics*, 18(9), 1094–1098.
<https://doi.org/10.1038/s41567-022-01662-4>



Turquet, A., Wuestefeld, A., Svendsen, G. K., Nyhammer, F. K., Nilsen, E. L., Persson, A. P.-O., & Refsum, V. (2024). Automated Snow Avalanche Monitoring and Alert System Using Distributed Acoustic Sensing in Norway. *GeoHazards*, 5(4), 1326–1345. <https://doi.org/10.3390/geohazards5040063>



Waves in Plates. (2014). In J. L. Rose (Ed.), *Ultrasonic Guided Waves in Solid Media* (pp. 76–106). Cambridge University Press.
<https://doi.org/10.1017/CBO9781107273610.008>



Zhu, L., & Rivera, L. A. (2002). A note on the dynamic and static displacements from a point source in multilayered media. *Geophysical Journal International*, 148(3), 619–627. <https://doi.org/10.1046/j.1365-246X.2002.01610.x>

MPM STF Workflow

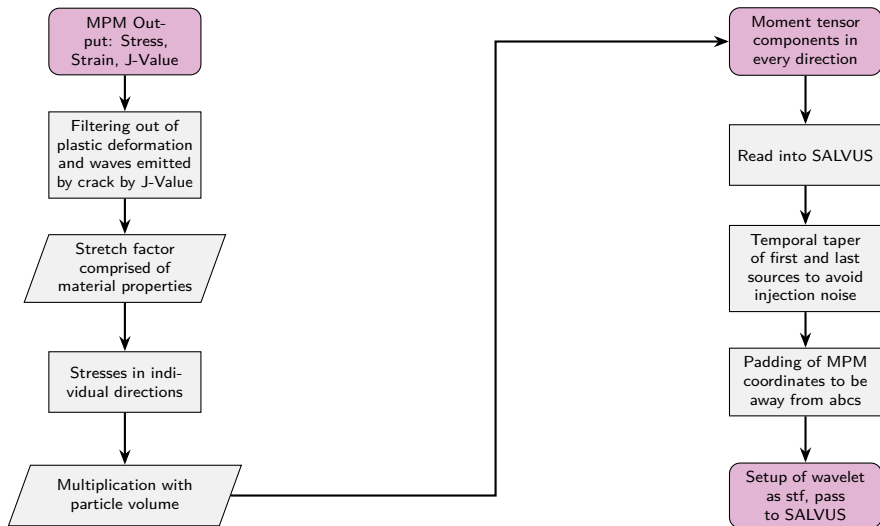


Figure: Workflow for conversion of MPM data and input as SALVUS STF.

Simple model velocity plots I

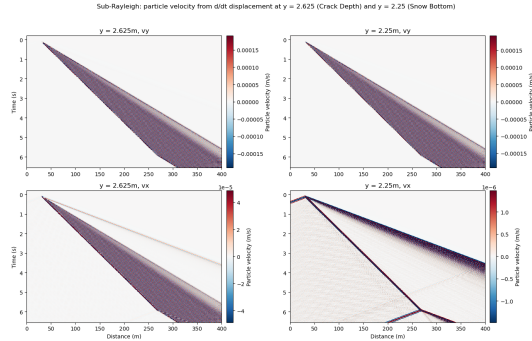


Figure: Simple model Sub-Rayleigh velocity plots. Crack depth on left side, snow bottom on right. v_y at top, v_x at bottom. Velocity derived from displacement.

Simple model velocity plots II

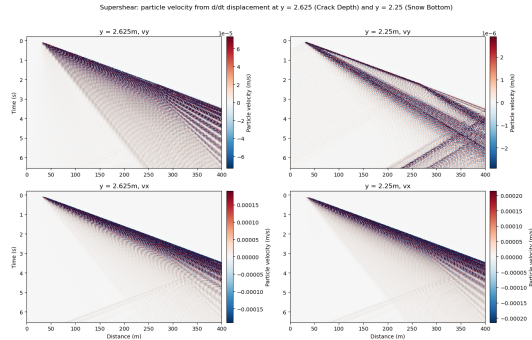


Figure: Simple model Supershear velocity plots. Crack depth on left side, snow bottom on right. v_y at top, v_x at bottom. Velocity derived from displacement.

MPM velocity plots

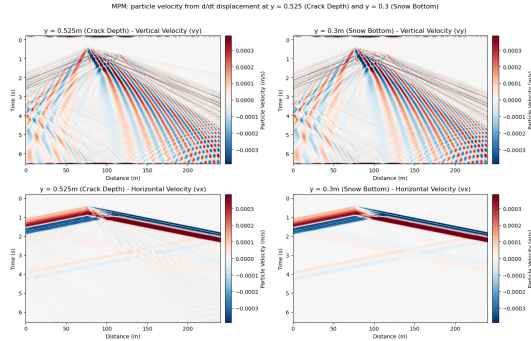


Figure: MPM-guided model velocity plots. Crack depth on left side, snow bottom on right. v_y at top, v_x at bottom. Velocity derived from displacement.

Simple model F-k plots and dispersion curves

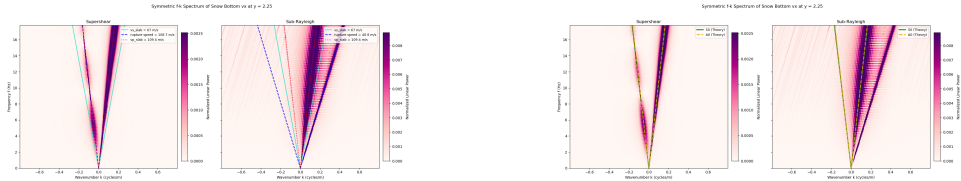


Figure: Supershear and Sub-Rayleigh F-k spectra with medium and Lamb-wave dispersion curves indicated by colored lines.

MPM F-k plots

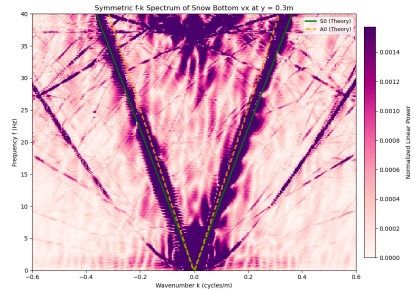
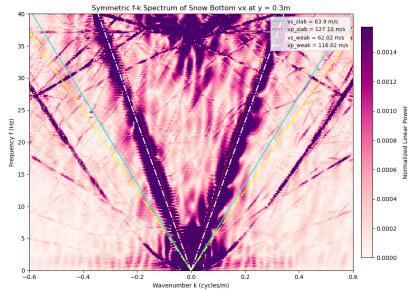


Figure: MPM F-k spectra with medium and Lamb-wave dispersion curves indicated by colored lines.

MPM STF cumulative Integral

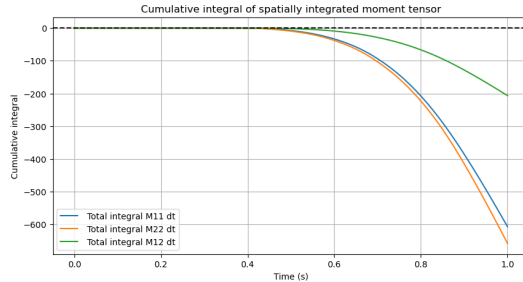


Figure: Cumulative integral over time of all MPM STF components



Salome Bachmann
Master Student
sbachmann@student.ethz.ch

ETH Zurich
D-EAPS

Temporary page!

$\text{\LaTeX}$ was unable to guess the total number of pages correctly. As there was some unprocessed data should have been added to the final page this extra page has been added to receive it.

If you rerun the document (without altering it) this surplus page will go away, because $\text{\LaTeX}$ now knows how many pages to expect for this document.